

Project Initialization and Planning Phase

Date	06 July 2026
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to assist farmers in selecting the most suitable crop using Artificial Intelligence and Machine Learning. The system analyzes soil nutrients, weather conditions, rainfall, temperature, humidity, and pH values to recommend the best crop and estimate yield. This helps farmers improve productivity, reduce risks, and increase profitability through data-driven decisions.

Project Overview	
Objective	To develop an AI-powered crop recommendation and yield prediction system that helps farmers choose the right crop based on soil and environmental conditions while improving agricultural productivity. The system recommends suitable crops using a trained Random Forest Machine Learning model and displays crop information through a Flask web application.
Scope	The project provides intelligent crop recommendations, predicts crop yield, analyzes environmental factors, and supports farmers in making informed cultivation decisions using machine learning.
Problem Statement	
Description	Farmers often select crops without analyzing soil quality and climatic conditions, resulting in low yield, financial loss, and inefficient resource utilization.
Impact	The proposed system improves crop selection accuracy, increases agricultural productivity, minimizes losses, optimizes resource usage, and supports sustainable farming practices.
Proposed Solution	
Approach	The system uses Machine Learning algorithms to analyze soil nutrients (N, P, K), pH, rainfall, temperature, and humidity to recommend suitable crops and predict expected yield.
Key Features	AI-based Crop Recommendation ,Soil Nutrient Analysis ,Crop Image Display ,Telugu Crop Name Translation ,Flask-Based Web Interface

	- Real-time analysis of soil and environmental conditions. - Accurate crop yield prediction using machine learning.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Intel Core i5 Processor (4 Cores)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Data Source, size, format	Kaggle Crop Recommendation Dataset, CSV (~2,200 records)